M.I.N.D.

Mental Interpretation Network for Decision Making

Converting noisy EEG intent into safer FIRE/HOLD control using a safety-constrained threshold + experimental debounced FIRE/HOLD layer.

Team: Ishani Singh, Mayu Kanai, Habiba Hisham, Kevin Zhou, Joshua Zhuravskiy, Mohammad Yamout
Mentor: Asan Abdulkareem

The Safety-Constrained Need

Unsafe false activations can cause harm in rehabilitation. Safety is a first-class requirement.

Real-World Context: In Jan 2024, Noland Arbaugh, the first Neuralink recipient, faced long-term thread retraction & reduced performance. **Takeaway: Long-term stability is hard.**

Condition	Wrong-fire behavior	Safe-fire objective	Stability behavior
Ensemble only (no safety gate)	Tends toward higher risk without explicit constraint	Not explicitly optimized under alpha	Associated with higher jitter risk
Ensemble + safety gate	Constrained by alpha	Aimed at increasing safe-fire under constraint	Associated with moderate jitter near threshold
Ensemble + debounced safety layer (Experimental) (alpha=0.05)	**Constrained by alpha**	**Aimed at increasing safe-fire under constraint**	Can reduce toggling risk via temporal smoothing (hysteresis, k-of-n, latch)

Designing for Real Usage

Optimizing for multiple priorities across disease pathways (ALS, Stroke, SCI).

1

Patients

Need reliable intent-to-action with minimal fatigue and risk.

2

Clinicians (OT/PT)

Need stable behavior, clear failure modes, safe defaults.

3

Caregivers

Need predictability and reduced risk of unintended movement.

4

Insurers

Need evidence of safety and value for broader adoption.

5

Low-Resource

Need lower-cost pathways & less reliance on specialists.

Literature & Framing

EEG is noisy (artifacts). Safety and stability are as important as raw accuracy.

Key Literature Basis

Methods:

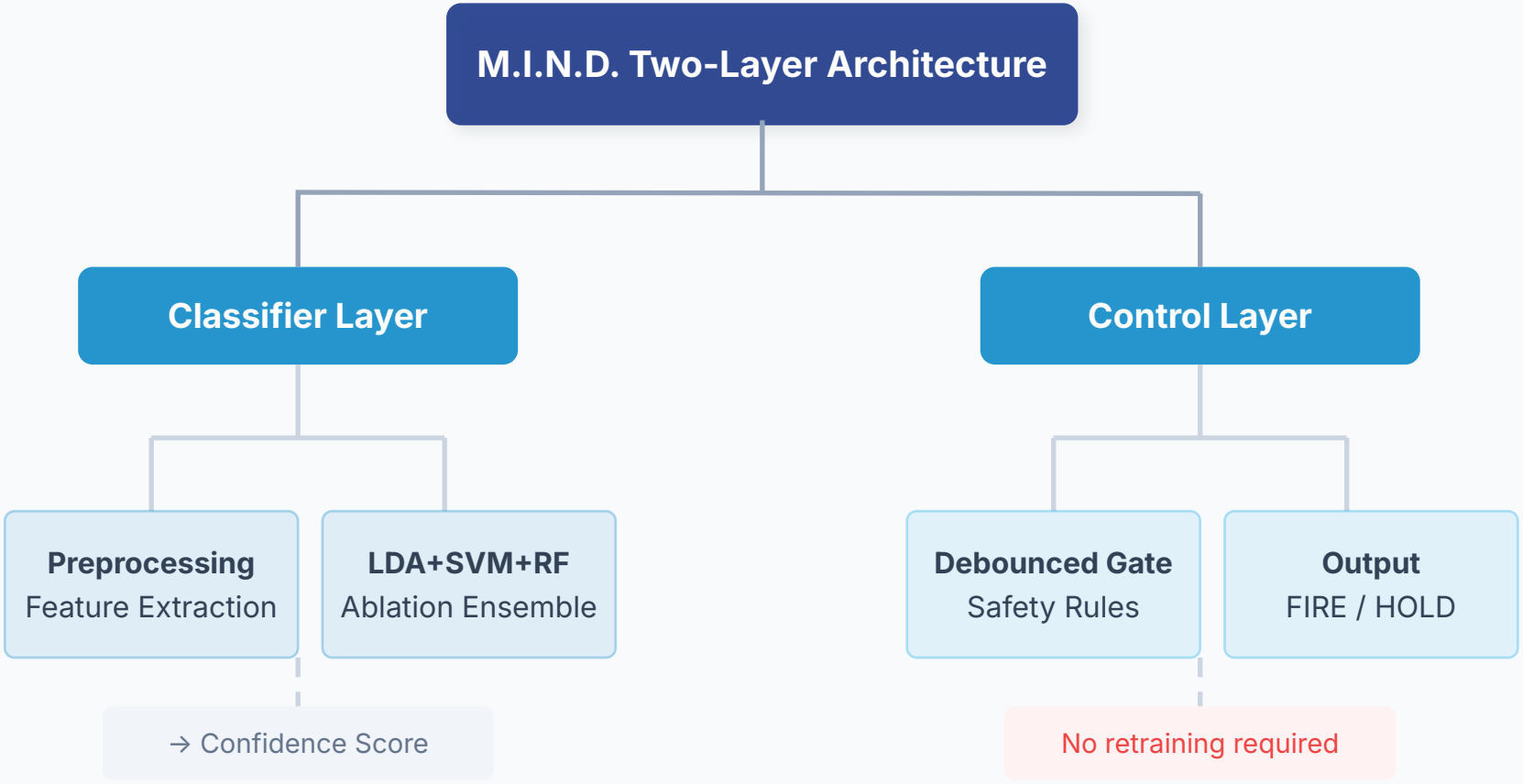
- CSP / FBCSP
- Ensemble RCSSP
- SVM Voting
- ICA Artifact Correction

Challenges Identified:

- Inefficient BCI users
- Low SNR & non-stationarity
- BCI latency

Datasets:

- BCI Competition IV 2a
- BCI Competition III
- PhysionetMI EEG motor-imagery resources
- BNCI Horizon 2020 Dataset BNCI2014-001 (4-class motor imagery EEG)



Hypotheses & Operating Policy

Distinguishing confirmatory tests from exploratory tuning thresholds.

Confirmatory Test

Locked Point t* = 0.60

Hypothesis 1:
Ensemble confident accuracy > per-subject best-single at locked threshold t* = 0.60.

Policy Impact:
Locked threshold supports fair, unbiased comparison across all models before inferential statistics.

Exploratory / Transparency

Sweep {0.55 - 0.75}

Hypothesis 2:
Ensemble reduces across-subject performance variance.

Policy Impact:
Evaluates the coverage vs. confident accuracy tradeoff. Reported even if not supported.

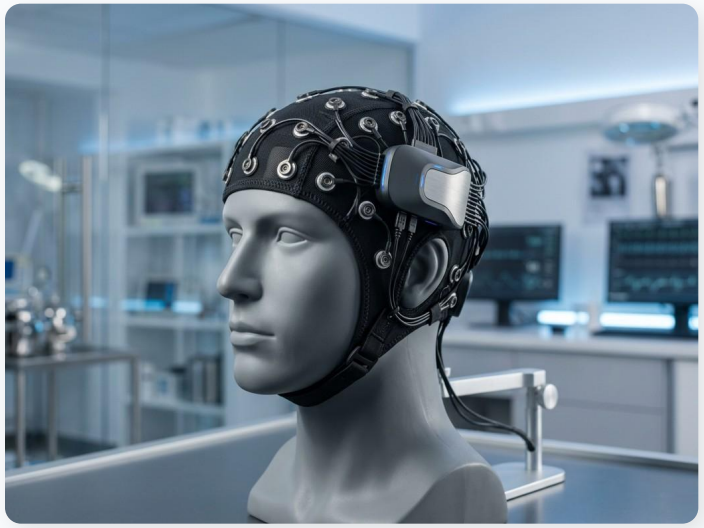
Signal to Model

Robust preprocessing and an ablation ensemble handle EEG noise to produce confidence scores.

1. Raw EEG
Artifact-prone signals including blinks, muscle, and motion artifacts.

2. Preprocessing
Robust artifact correction and feature extraction.

3. Ablation Ensemble
LDA + SVM + RF with global soft voting → Outputs Confidence Score.



Improving Stability Across Noisy Trials
Because EEG is non-invasive, it introduces high noise levels. We use robust feature steps paired with a multi-model ensemble to smooth out variations.
The Outcome: Rather than forcing a single classification, the ensemble outputs a continuous probability (Confidence Score) that feeds into the safety layer.

Policy to Action

A safety-first control layer wraps the classifier without requiring retraining.

1. Confidence Score (from Ensemble)

↓

2. Threshold Evaluation ($t^* = 0.60$)

↓

3. Debounced Safety Gate

↓

4. FIRE / HOLD Output

Safety Rule:
Prefer HOLD over unsafe movement when confidence is low or unstable. By wrapping the classifier in policy, we enforce safety without requiring a new model.

Rigorous Evaluation Protocol

Data splits, leakage prevention, and explicit inferential boundaries.

Data & Protocol

BCI Comp IV 2a
9 subjects tested under within-subject 5-fold CV.

Leakage Prevention
Each subject partitioned independently. No trials share training & evaluation sets.

Key Metrics

Coverage
Fraction of trials above threshold (acted upon).

Confident Accuracy
Accuracy ONLY on committed trials.

Kappa
0.550 global agreement.

Guardrails

No Post-Hoc Tuning
Confirmatory threshold locked at $t^* = 0.60$.

Transparency First
Exploratory thresholds and failed stability tests are reported transparently.

Model Behavior Tradeoffs

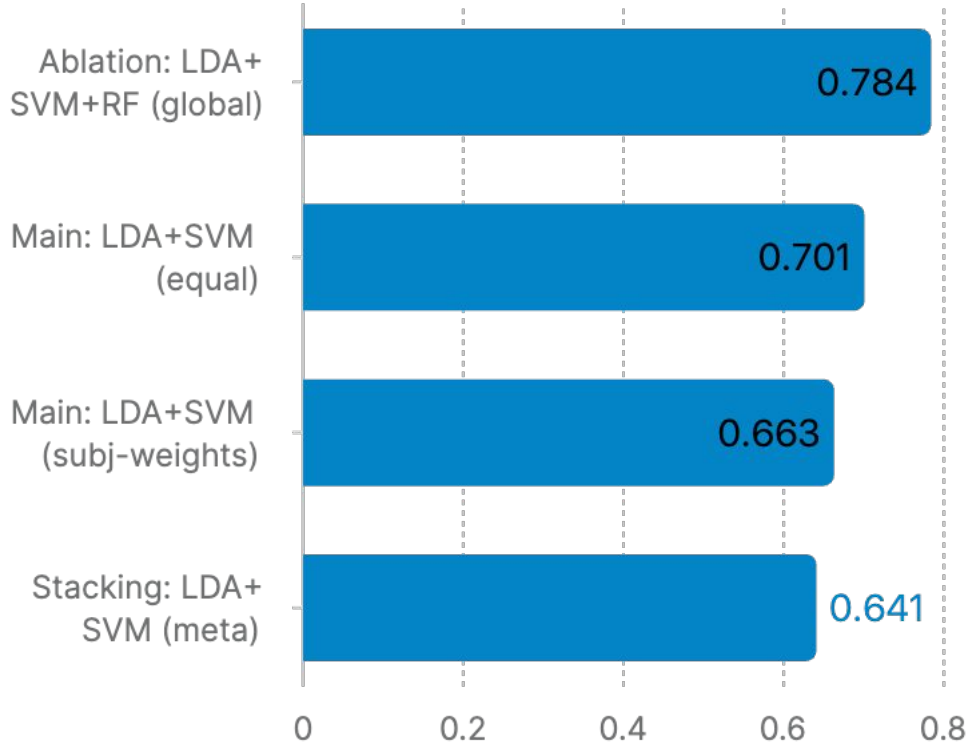
Operating regimes at locked threshold t* = 0.60.

Main: LDA+SVM (subj-weights)

- **Coverage:** 81.6% (High Coverage)
- **Conf. Accuracy:** 0.663
- **Delta vs Baseline:** +0.035 (p=0.024)

Ablation: LDA+SVM+RF (global)

- **Coverage:** 46.4% (Lower)
- **Conf. Accuracy:** 0.784
- **Delta vs Baseline:** +0.156 (p=0.00016)



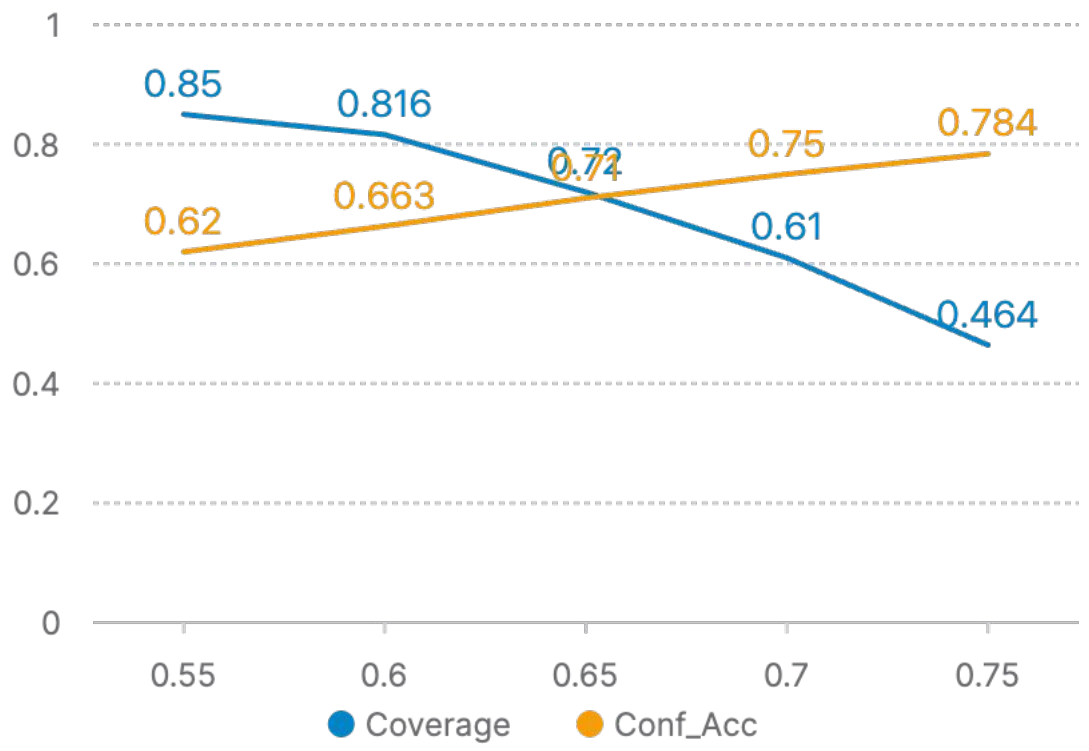
Accuracy vs. Coverage (Sweep)

Evaluating the exploratory sweep across thresholds {0.55 - 0.75}.

Trading Coverage for Confidence.

As the safety threshold increases, confident accuracy rises steadily, but coverage drops.

The safety gate forces the system to prefer **HOLD** over uncertain **FIRE** commands, improving stability at the cost of actuation volume.



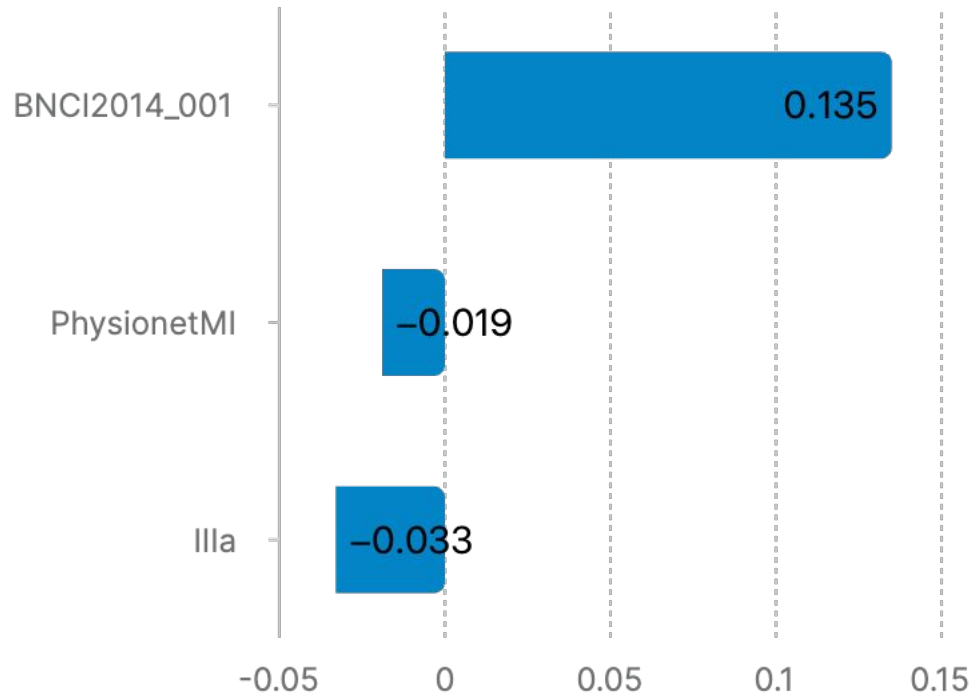
Validating the Locked Policy

Evaluating transfer performance across datasets at t* = 0.60.

BNCI2014_001 (n=10)
Strong Signal: Delta = +0.135 (p=0.004)
Displays significant confident accuracy lift.

PhysionetMI (n=10)
Limitation: Delta = -0.019 (p=0.167)
Dataset shift impedes transfer of locked policy.

IIIa (n=3)
Underpowered: Delta = -0.033 (p=0.183)
Sample size is too small for significance.



The Safety-Constrained Innovation

Aiming to increasing safe-fire subject to wrong-fire ≤ alpha (0.05).

Existing BCI Approaches		Our Approach (M.I.N.D.)
Classifier-only output; always commits to the top class	Output	Safety-constrained threshold + **FIRE/HOLD policy** in experimental debounced Gate; favors HOLD on low confidence
Optimized for headline accuracy	Objective	**Aimed at increasing safe-fire** under constraint
Model-specific safety logic	Scope	**Model-agnostic** (no retraining)

Hysteresis & Debouncing

Controller mechanics and experimental safety-feature tuning.

Debounced Controller Logic

Hysteresis

t_on > t_off, preventing oscillation around the threshold.

k-of-n Confirmation

Fires only if confidence is sustained over multiple frames.

Latch-While-Firing

Reduces rapid class flipping during active movements.

Experimental Settings (Transparency Report)

Model	t_on/off	k-of-n	Coverage	Wrong-fire	Safe-fire	Toggle
LDA_SVM (subj-weights)	0.75/0.65	4-of-5	2.4%	1.7%	0.6%	0.007
LDA_SVM_RF (global)	0.60/0.50	4-of-5	3.9%	2.8%	1.1%	0.009

Interpretation: These settings met the wrong-fire ≤ 0.05 safety constraint but were too conservative for usability. **Next step:** Tune to increase safe-fire and coverage.

Designing for Equity

Priorities, metrics, and scale guardrails from a human-centered design perspective.

Human-Centered Design

Rehab Priorities

- **Safety First:** Fewer unsafe activations
- Low fatigue & predictable behavior
- Practical setup time
- Policy: HOLD > unsafe movement

Sustainability Metrics

- **Wrong-fire parity:** $|gap| \leq 0.02$
- **Safe-fire parity:** $|gap| \leq 0.05$
- Cost ceiling trend decreases each phase

Scale Guardrails

- **Phase Gate Reviews:** Safety & fairness sign-off
- Public site summaries
- Strict incident logs
- Clinician-supervised starts

Pathway to Deployment

From OpenBCI to Raspberry Pi to FES: A phased approach to clinical integration.

Phase 1: Bench Pilot
(4-6 weeks)

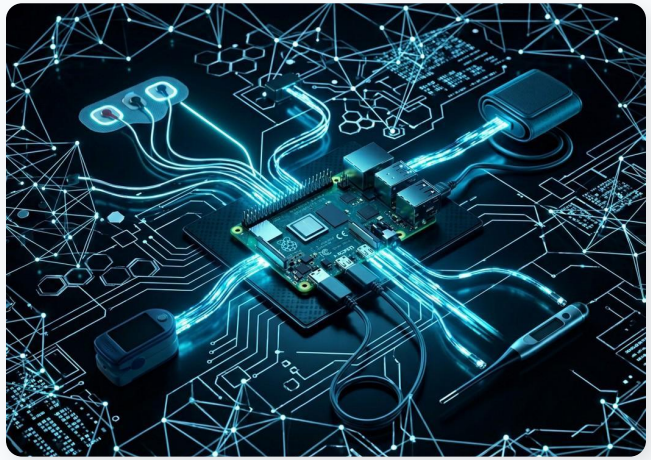
Latency, wrong-fire stability, simulated actuators

Phase 2: Supervised Sim
(6-8 weeks)

Coverage usability & safety adherence

Phase 3: Multi-site
(8-12 weeks)

Cross-site reproducibility & fairness spread



Cost Model & Commercial Viability

- **Prototype Stack:** $715-$900 total (driven largely by Ganglion EEG cost)
- **Clinic Pilot Cost:** ~$15-$40/session (equipment amortization + consumables)
- **Cost-Down Path:** Move toward sub-$500 using lower-cost EEG options, shared devices, and setup automation.

Deployable with existing components; the main remaining step is reliability validation, not a new core invention.

Scientific Integrity & Limits

Explicit constraints on the current findings. We do not overclaim.

No Patient Trials
Current results come from regulated benchmark datasets, not real patients yet.

Dataset Shift
Transfer performance is reduced under the locked policy (e.g., PhysionetMI).

EEG Noise
Blinks, muscle, and motion artifacts remain real-world robustness hurdles.

Stability Variance
No significant variance reduction at locked $t^*=0.60$ (Levene $p=0.568$).

Small Sample Sizes
IIIa has only $n=3$, making hypothesis tests underpowered.

Actionable Roadmap

Immediate steps for hardware integration and clinician validation.

1. Hardware Integration
Latency tests mapping OpenBCI to Pi to actuator.

2. Stress Testing
Non-stationarity stress testing simulating real-life EEG shifts.

3. Clinician Study
Feedback call (May 6) to define acceptable policy bounds.

4. Fairness Reporting
Per-subject coverage and abstention reporting for equity targeting.

5. Deploy Package
Low-cost deploy package: parts list, setup guide, reduced compute pathway.

Judge Summary

A model-agnostic, safety-constrained BCI layer.

Problem

Unsafe false activations can make assistive BCI control clinically risky. Rehabilitation access is constrained by high device costs, setup burden, and safety fears.

Innovation

A model-agnostic, safety-constrained threshold + experimental debounced FIRE/HOLD layer wrapped around an LDA+SVM+RF ablation ensemble. It optimizes for dependability under noise.

Evidence

At locked t*=0.60: Ablation model gives strong lift (**delta +0.156, p=0.00016**), with the main LDA + SVM model still providing **p=0.024** with **coverage=81.6%**. Policy computes fast (1.324 µs/trial in software-side policy compute time). Note: Variance reduction was not statistically significant (p=0.568).

The Ask

We are seeking support to advance to a **hardware-in-loop pilot** and to conduct clinician co-design workshops to finalize our deployment policy and fairness safeguards.

M.I.N.D.

Problem: Unsafe false activations harm users and reduce trust.

Innovation: A model-agnostic, safety-constrained threshold with a FIRE/HOLD layer + experimental debounced layer.

Evidence: Delta +0.156 ($p=0.00016$) at locked threshold.

Ask: Support a supervised hardware-in-loop pilot and clinician co-design.

Thank you.